Physics-Informed Deep U-Fourier Neural Network for Efficient Prediction of Guided Wavefield in Plates with Irregular Defects

Zhao Wang[1], Linfeng Wang[1], Xiao Ying[1], Jin Hou[2], and Yang Liu[1,*]

[1] *State Key Laboratory of Precision Measuring Technology and Instruments, Tianjin University, Tianjin, 300072, China*

[2] *CNOOC Research Institute Company Limited, Beijing, 100028, China*

[*] Corresponding author.
E-mail address: ultrasonicslab@tju.edu.cn (Y. Liu)

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Abstract

Accurate prediction of the full-field ultrasonic guided wave evolution in plates with irregular defects is essential for quantitative nondestructive evaluation(QNDE). Traditional numerical solutions encounter efficiency bottlenecks while existing data-driven models are mostly limited by simplified scalar acoustic equations which struggle to characterize dispersion effects in plates with variable thickness. This paper proposes a physics-informed deep U-Fourier neural network (PI-DeepUFNet) merging Kirchhoff-Love (KL) plate theory with neural operators for the efficient prediction of spatiotemporal wavefield in thin plates containing irregular defects. The framework combines the local multi-scale geometric feature extraction advantages of U-Net with the global spatiotemporal evolution modeling capability of 3D Fourier Neural Operators (FNO). By discretizing the fourth-order flexural wave equation into finite-difference operator and explicitly embedding them into the loss function, joint optimization of data and physics is achieved. Numerical experiments show that the model accurately predicts the A0 mode defect scattering response under complex corrosion conditions. The average RMSE, RL2, and PCC for wavefield prediction on the test set reach 3.08×10^{-3}, 6.98×10^{-2}, and 0.9971, respectively. Ablation experiments confirm that physical constraints effectively enhance the prediction stability and dynamical consistency of the model in complex scattering scenarios. Geometric sensitivity analysis and scanning laser Doppler vibrometer (SLDV) experiments further verify its robustness in real physical environments. The forward modeling paradigm established in this study possesses good theoretical portability and

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provides a universal approach for the deep integration of physical priors with operator learning.

Keywords: ultrasonic guided wave, wavefield prediction, physics-informed neural operator, KL plate theory, defect scattering

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1. Introduction

Plate-like structures are widely used in modern industrial fields represented by aerospace, shipbuilding, and petrochemical engineering, with their structural integrity directly linked to operational safety and economic efficiency. Owing to the advantages of long propagation distance, low attenuation, and high sensitivity to local thickness perturbations, ultrasonic guided wave (UGW) has emerged as a crucial information carrier in the nondestructive testing (NDT) and structural health monitoring (SHM) of these components [1, 2]. The interaction between guided waves and irregular defects within a plate induces complex scattering phenomena, including reflection, transmission, and diffraction [3]. The accurate solution of this full-field forward problem, involving multiple scattering and dispersion evolution dynamics, aims to establish a deterministic mapping from defect geometric distributions to dynamic wavefield responses. This task represents a fundamental physical challenge in understanding guided wave scattering mechanisms and serves as an indispensable computational foundation for subsequent quantitative inversion and high-resolution imaging [4, 5].

Existing research on guided wave propagation and scattering is generally categorized into three types: analytical, numerical, and data-driven methods [6]. Analytical methods are commonly based on Green's function methods, Helmholtz integral equations, or generalized Bessel function expansions, are dedicated to mathematically deriving exact closed-form solutions for scattered wavefield [7, 8]. However, analytical solutions rely on regular geometric boundaries and simplified

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physical assumptions, making it difficult to directly handle full-wavefield scattering problems involving irregular defects [9, 10]. To overcome geometric topological constraints, numerical methods represented by the finite element method (FEM), finite difference method (FDM), and boundary element method (BEM) have become the mainstream standard for addressing guided wave propagation under complex boundary conditions [11, 12, 13]. By finely discretizing continuous partial differential equation (PDE) domains in space and time, numerical methods are capable of approximating complex physical boundaries and material inhomogeneities with high accuracy. Moreau et al. established a high-accuracy finite element scattering evolution model for irregular corrosion defects in plate structures [14]. Nguyen et al. utilized numerical algorithms to solve dynamic acoustic fields under different boundary conditions and analyzed their stability and accuracy [15]. Nevertheless, to ensure numerical stability for high-frequency wave motions, conventional numerical solvers still encounter significant computational efficiency bottlenecks in scenarios involving large-scale real-time wavefield prediction or extensive forward simulations [16, 17].

To overcome the efficiency bottlenecks of iterative calculations in numerical methods, data-driven deep learning techniques have been introduced in recent years for forward prediction of physical fields [18, 19, 20]. Zhang et al. proposed a hybrid prediction model merging convolutional neural networks and Transformers to predict far-field scattering signals in thin plates with defects [21]. Wang et al. developed a fusion model based on compressive sensing and deep learning, which achieved high-fidelity reconstruction of guided wave signals under sparse sampling conditions and

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successfully performed defect imaging using the reconstructed data [22, 23]. Although neural networks have demonstrated significant potential in fitting nonlinear mappings of complex systems, conventional purely data-driven models, due to their inherent black-box nature, not only depend heavily on massive amounts of high-quality training data, but also tend to generate non-physical artifacts in the absence of prior constraints [24, 25, 26]. Raissi et al. proposed physics-informed neural networks (PINNs), which substantially improved physical consistency by embedding PDEs into the loss function as regularization residuals [27]. Rasht-Behesht et al. combined PINNs with acoustic wave equations and successfully addressed seismic wave propagation and full waveform inversion (FWI) problems [28]. To break through the limitations of specific samples and grid resolutions, Lu et al. proposed the deep operator network (DeepONet) based on the universal operator approximation theorem, achieving continuous mapping in infinite-dimensional function spaces [29]. Subsequently, Wang et al. further proposed the physics-informed DeepONet, which achieved efficient learning of general solution operators for PDEs by combining automatic differentiation (AD) to calculate physical residuals [30]. In acoustics, Nair et al. combined the physics-informed operator network with non-rational B-splines geometric parameterization to realize the forward prediction of acoustic scattering fields for arbitrary scatterers [31]. Wang et al. demonstrated the effectiveness of physical priors through a series of progressive studies and proposed embedding discrete finite-difference operators into the network as an alternative to AD. This approach effectively circumvented computational memory bottlenecks and successfully predicted transcranial frequency-domain wavefield as well

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as S0 mode Lamb wave scattering in plate structures [32, 33, 34, 35]. Meanwhile, another class of architectures represented by the Fourier neural operator (FNO) has made breakthrough progress in complex spatiotemporal evolution simulations, owing to its advantage in capturing global correlations in the frequency-domain [36, 37].

Despite the widespread application of deep learning to wavefield problems, existing research on guided waves primarily focuses on inverse problems, with a noticeable scarcity of studies targeting the rapid forward prediction of full spatiotemporal wavefield under complex corrosion-like defects [38, 39]. Furthermore, most current models rely on simplified scalar acoustic wave equations, failing to effectively characterize the dispersion effects and local stiffness discontinuities inherent in plate waves [40]. Constrained by the computational bottlenecks associated with high-order AD during full-field training, a high-resolution spatiotemporal solution framework that simultaneously accounts for realistic dispersion evolution and complex geometric coupling under explicit physical constraints remains unavailable.

Against this background, this paper proposes the physics-informed deep U-Fourier neural network (PI-DeepUFNet), a novel neural operator framework for the fast prediction of full guided wavefield in plates with irregular defects. By incorporating the fourth-order dynamic equations from Kirchhoff-Love (KL) plate theory as physical priors, the proposed framework integrates the local geometric feature extraction capabilities of U-Net with the global spatiotemporal evolution modeling of 3D FNO. This synergy enables efficient and physics-consistent full spatiotemporal prediction of ultrasonic guided wavefield in variable thickness plates.

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The main contributions of this work are as follows:

(1) KL plate theory is employed as an alternative to simplified acoustic models to construct a finite-difference forward operator for flexural waves under spatially variable thickness conditions, and a dataset containing different scattering characteristics is generated.

(2) The PI-DeepUFNet architecture is designed to integrate local geometric feature extraction with global spatiotemporal modeling, while explicitly embedding discrete physical operators to achieve the joint optimization of data and physics.

(3) The performance and engineering utility of the proposed method in complex irregular defect scenarios are systematically evaluated through numerical simulations, ablation studies, and experimental validation.

The remainder of this paper is organized as follows: Section 2 delineates the mathematical model of guided wave propagation based on KL plate theory and its dispersion characteristics. Section 3 introduces the discrete forward modeling method, the PI-DeepUFNet architecture, and the design of the physics-informed loss function. Section 4 analyzes the convergence characteristics of model training and wavefield prediction results, with the role of physical constraints evaluated via ablation experiments. Section 5 discusses the sensitivity of the model to various defect morphologies and validates its applicability in real-world scenarios through scanning laser Doppler vibrometer (SLDV) experiments. Section 6 concludes the paper.

2. Theory

This section derives the dynamic partial differential governing equation under

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spatially variable thickness conditions based on KL thin plate theory. This variable-coefficient equation serves as the mathematical foundation for guided wave propagation in complex defective plates and will be embedded as prior physical knowledge into the loss function of the deep neural operator in subsequent sections.

2.1 Dispersion characteristics

KL theory describes the flexural behavior of isotropic thin plates under transverse loading, where the thickness is significantly smaller than the in-plane dimensions [41, 42]. Consider a thin plate with a spatially varying thickness $h(x, y)$, where the undeformed mid-surface is set at $z = 0$ and the out-of-plane displacement is denoted by $w(x, y, t)$, as shown in Fig. 1(a). Based on the Kirchhoff hypothesis, normals to the mid-surface remain straight and perpendicular to it after deformation, and transverse shear deformation is negligible. Consequently, under small-deflection conditions, the in-plane strains induced by mid-surface curvature are linearly distributed along the thickness direction. Accordingly, the normal strains of the plate can be expressed as:

$$\varepsilon_{xx}(x, y, z, t) = -z\frac{\partial^2 w}{\partial x^2}, \quad \varepsilon_{yy}(x, y, z, t) = -z\frac{\partial^2 w}{\partial y^2} \tag{2}$$

and the in-plane shear strain is:

$$\gamma_{xy}(x, y, z, t) = -2z\frac{\partial^2 w}{\partial x \partial y} \tag{3}$$

It can be seen that the strains vary linearly with the distance from the mid-surface, reaching equal and opposite values at the top and bottom surfaces. This corresponds to the upper and lower layers being under tension and compression, respectively, during plate bending. Under the plane stress assumption, the isotropic linear elastic

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constitutive relationship is given by:

$$\sigma_{xx} = \frac{E}{1-\nu^2}(\varepsilon_{xx} + \nu\varepsilon_{yy}), \quad \sigma_{xx} = \frac{E}{1-\nu^2}(\varepsilon_{yy} + \nu\varepsilon_{xx}), \quad \tau_{xy} = G\gamma_{xy} \tag{4}$$

where, E is Young's modulus, ν is Poisson's ratio, and G is the shear modulus.

By integrating the stresses along the thickness direction with respect to the mid-surface, the bending moments per unit width are defined as:

$$M_{xx} = \int_{-h/2}^{h/2} z\sigma_{xx}\mathrm{d}z, \quad M_{yy}(x,y,t) = \int_{-h/2}^{h/2} z\sigma_{yy}\mathrm{d}z, \quad M_{xy}(x,y,t) = \int_{-h/2}^{h/2} z\tau_{xy}\mathrm{d}z \tag{5}$$

Introducing the local flexural rigidity of the plate:

$$D(x,y) = \frac{E\left[h(x,y)\right]^3}{12(1-\nu^2)} \tag{6}$$

the relations between bending moments and curvatures can then be expressed as:

$$\begin{aligned} M_{xx}(x,y,t) &= -D(x,y)\left(\frac{\partial^2 w}{\partial x^2} + \nu\frac{\partial^2 w}{\partial y^2}\right) \\ M_{yy}(x,y,t) &= -D(x,y)\left(\frac{\partial^2 w}{\partial y^2} + \nu\frac{\partial^2 w}{\partial x^2}\right) \\ M_{xy}(x,y,t) &= -D(x,y)(1-\nu)\frac{\partial^2 w}{\partial x \partial y} \end{aligned} \tag{7}$$

Newton's second law governs the force equilibrium of the plate element. As shown in Fig. 1(b), the Kirchhoff shear forces are defined as:

$$Q_x = \frac{\partial M_{xx}}{\partial x} + \frac{\partial M_{xy}}{\partial y}, \quad Q_y = \frac{\partial M_{yy}}{\partial y} + \frac{\partial M_{xy}}{\partial x} \tag{8}$$

Considering the inertial force and external load $q(x,y,t)$, the equilibrium equation for the micro-element can be written as:

$$\frac{\partial Q_x}{\partial x} + \frac{\partial Q_y}{\partial y} + q(x,y,t) = \rho h(x,y)\frac{\partial^2 w}{\partial t^2} \tag{9}$$

where, ρ represents the material density. Substituting Eq. (7) yields the equilibrium equation in terms of moments:

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$$\frac{\partial^2 M_{xx}}{\partial x^2}+2\frac{\partial^2 M_{xy}}{\partial x\partial y}+\frac{\partial^2 M_{yy}}{\partial y^2}+q(x,y,t)=\rho h(x,y)\frac{\partial^2 w}{\partial t^2} \tag{10}$$

Further substitution of Eq. (6) yields the KL dynamic governing equation for thin plates with variable thickness undergoing out-of-plane vibrations at small deflections. For general cases involving variable stiffness, the compact form can be expressed as:

$$\nabla^2(D\nabla^2 w)-(1-\nu)\Diamond(D,w)+\rho h\frac{\partial^2 w}{\partial t^2}=q \tag{11}$$

where, $\nabla^2(D\nabla^2 w)$ is the primary biharmonic term encompassing the low-order effects of stiffness variation, and $\Diamond(D,w)$ denotes the mixed operator term introduced by the stiffness gradient. The third term represents the inertial component. This equation explicitly demonstrates that irregular thickness variations introduce additional source terms into the equation, which constitutes the physical origin of scattering wave generation.

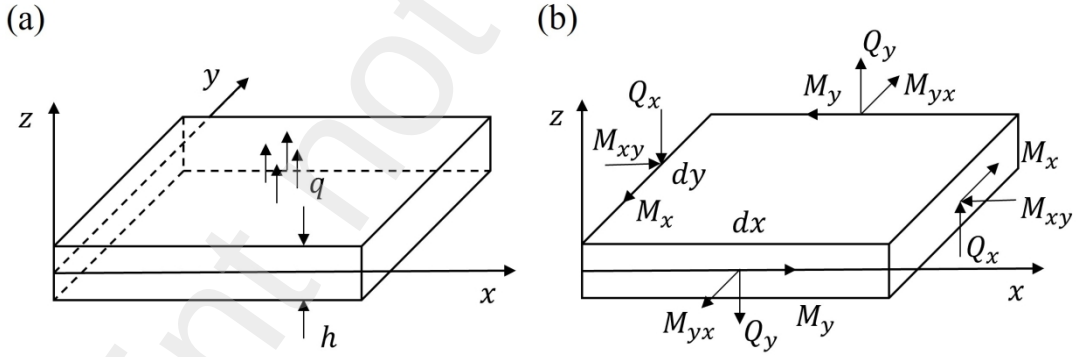


Fig. 1. Schematic of the KL thin plate model. (a) Geometric configuration and coordinate system, (b) Bending moments and Kirchhoff shear forces on a plate element.

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2.2 Dispersion relation

To quantitatively analyze the influence of thickness variations on guided wave propagation, a local homogenization treatment is applied to the previously mentioned KL governing equations. By assuming the thickness h and flexural rigidity D are constant within a local element and neglecting external forces, the governing equation simplifies to:

$$D\frac{\partial^4 w}{\partial x^4}+\rho h\frac{\partial^2 w}{\partial t^2}=0 \tag{12}$$

Under the assumption of thin plates at low frequencies, this fourth-order partial differential equation accurately represents the asymptotic behavior of the $\mathrm{A0}$ Lamb wave mode and inherently captures the dispersion characteristics from the underlying physical mechanism.

Substituting the plane harmonic wave solution $w(x,t)=Ae^{i(kx-\omega t)}$ yields the dispersion equation:

$$Dk^4-\rho h\omega^2=0 \tag{13}$$

where, k is the wavenumber and ω is the angular frequency.

Equation (12) establishes a nonlinear relationship between angular frequency and wavenumber, specifically $\omega \propto k^2$. Consequently, the phase velocity c_p and group velocity c_g can be derived as:

$$c_p=\frac{\omega}{k}=\sqrt{\frac{D}{\rho h}}k,\quad c_g=\frac{d\omega}{dk}=2\sqrt{\frac{D}{\rho h}}k \tag{14}$$

These expressions indicate that wave propagation speed increases with frequency. Specifically, higher frequency components travel faster, which leads to gradual pulse

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broadening and waveform distortion of broadband excitation signals during propagation, manifesting as clear dispersion effects.

Table 1
Material parameters.

ρ	E	ν	h
(kg/m^3)	(GPa)		(mm)
2700	70	0.33	3

The dispersion curves for phase and group velocities, generated by substituting the aluminum plate parameters from Table 1, are shown in Fig. 2. The normalized spectrum of the excitation signal is overlaid as a green dashed line to define the velocity range within the effective bandwidth.

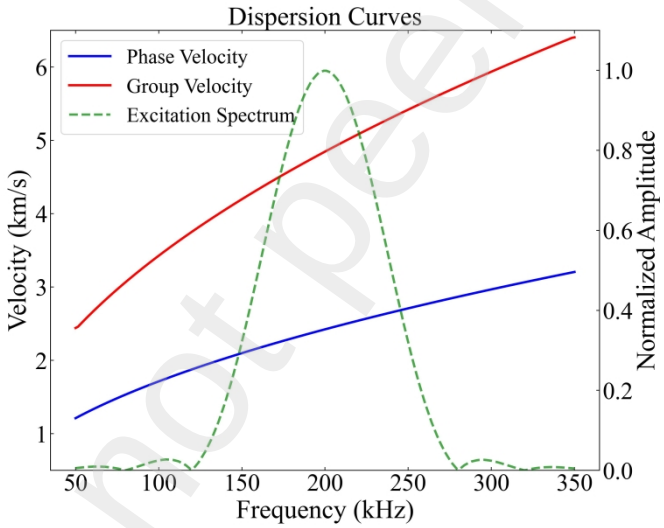


Fig. 2. Theoretical dispersion curves of the $A0$ Lamb wave mode for a 3 mm thick aluminum plate based on KL theory.

In contrast to simplified second-order acoustic wave models, KL plate theory incorporates fourth-order spatial derivative terms to precisely capture dispersion effects arising from flexural rigidity. Since the broadband propagation of guided waves in thin-walled media is inevitably accompanied by pulse broadening and distortion, non-dispersive models that ignore these effects are inadequate for accurate full-wavefield prediction in complex spatiotemporal domains. This derivation clarifies the dispersion

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laws of guided waves in plates and underscores the necessity of integrating this physical mechanism into wavefield prediction models.

3. Methodology

The previous section established the governing equations for flexural guided waves in plates with variable thickness based on linear elastodynamics and KL plate theory. This section further defines the wavefield solution problem under conditions of irregular defects, introduces the numerical forward operator and the PI-DeepUFNet framework employed for the solution.

3.1 Problem Formulation and Numerical Forward Operator

For thin plate structures containing irregular defects, the propagation of A0 mode Lamb waves can be regarded as a scattering problem described by a variable-coefficient biharmonic equation. Let $\Omega \subset \mathbf{R}^2$ be the two-dimensional spatial domain of the plate, where $\mathbf{x} = (x, y)$ denotes the spatial coordinates. Irregular defects are modeled geometrically through a thickness field $h(\mathbf{x})$ that varies continuously in space. Non-uniform variations in thickness at the defect locations simultaneously induce changes in local mass distribution and flexural rigidity, thereby acting as equivalent secondary scattering sources during the propagation of guided waves.

Within this physical framework, the total out-of-plane displacement wavefield $w(\mathbf{x}, t)$ can be decomposed into the superposition of the incident and scattered wavefield:

$$w(\mathbf{x}, t) = w_{inc}(\mathbf{x}, t) + w_{scat}(\mathbf{x}, t) \tag{15}$$

where, w_{inc} represents the reference incident wavefield propagating in a uniform thin

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plate without defects, and w_{scat} is the scattered wavefield excited by the thickness perturbation $\Delta h(\mathbf{x})$. The primary objective of this study is to construct a nonlinear mapping operator G_θ that, through optimization of parameters θ, establishes an end-to-end mapping from the two-dimensional geometric distribution of defects to the three-dimensional full spatiotemporal wavefield:

$$\hat{w}(\mathbf{x},t) = \mathsf{G}_\theta(h(\mathbf{x}), w(\mathbf{x},t_0)) \tag{16}$$

Training the deep neural operator G_θ requires the construction of a high-quality dataset containing diverse irregular defects and complex scattering features. Since traditional analytical solutions are typically restricted to simple boundaries, such as circles or rectangles, they are inadequate for handling natural corrosion defects with arbitrary shapes. Consequently, this study adopts the FDM to build a high-fidelity numerical forward operator [12]. Considering that the KL governing equation involves complex spatial derivatives of the fourth-order, direct discretization of the fourth-order operator would lead to an excessively large calculation template and reduced numerical stability. To address this, we rewrite the original partial differential equation into a coupled form of second-order operators for bending moment calculation and dynamic equilibrium. At a discrete time step t_n, the bending moment field $\boldsymbol{M}^n(\mathbf{x})$ corresponding to the spatial displacement field $w^n(\mathbf{x})$ is calculated as:

$$\boldsymbol{M}^n(\mathbf{x}) = -D(\mathbf{x}) \odot \left[\mathbf{L} * w^n(\mathbf{x}) \right]_v \tag{17}$$

where, $D(\mathbf{x})$ is the distribution of flexural rigidity determined from the thickness field $h(\mathbf{x})$, $\odot$ denotes the Hadamard product, and $*$ represents the convolution operation. $\mathbf{L}$ is the second-order spatial difference operator, and the subscript $(\cdot)_v$ indicates a

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linear combination that accounts for the Poisson's ratio coupling term. By utilizing the spatial divergence of the bending moment field, the current out-of-plane acceleration field $a^n(\mathbf{x})$ can be further determined:

$$a^n(\mathbf{x}) = \frac{1}{\rho h(\mathbf{x})} \odot \left[\mathbf{L} * \boldsymbol{M}^n(\mathbf{x}) \right] \tag{18}$$

Equations (16) and (17) facilitate the construction of a complete discrete differential operator, that describes the transient dynamic behavior of flexural guided waves in plates with variable thickness. In the temporal direction, explicit time integration is performed using a central difference scheme. The wavefield evolution equation is expressed as:

$$w^{n+1} = 2w^n - w^{n-1} + \Delta t^2 \cdot \mathsf{K}_{KL}(w^n, h) + S^n \tag{19}$$

where, S^n is the external excitation source term in discrete form. This explicit finite difference time-domain (FDTD) scheme offers advantages including algorithmic simplicity and high efficiency in parallel computing. With appropriate grid partitioning, it precisely captures subtle scattering details of guided waves induced by irregular thickness variations. To avoid non-physical numerical dispersion and ensure the convergence of the explicit integration, the time step Δt and the spatial grid spacing Δx must strictly satisfy the Courant-Friedrichs-Lewy (CFL) stability constraint. This ensures that the physical propagation distance of the wave within a single time step does not exceed the spatial grid spacing [17].

Provided that the initial and boundary conditions along with material parameters are fixed, the complete spatiotemporal evolution of the wavefield is uniquely determined by the geometric thickness distribution of the defects $h(\mathbf{x})$. The numerical

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forward physical model constructed by the FDTD scheme can be formalized as a deterministic nonlinear operator:

$$\mathbf{w} = \mathsf{F}\ (h) \tag{20}$$

This operator provides accurate supervisory signals for the subsequent learning of the deep neural operator. By performing parallel forward calculations on multiple sets of randomly generated irregular thickness fields, a guided wave wavefield dataset rich in diverse scattering patterns is efficiently produced. This dataset establishes the foundation for the training and validation of the PI-DeepUFNet framework.

3.2 Forward modeling of helical wave propagation

To address the computational efficiency and accuracy bottlenecks inherent in the forward modeling of guided waves in plates with variable thickness, we develop the physics-informed deep U-Fourier neural network. This framework integrates the advantages of U-Net in local geometric feature extraction across multiple scales with the capacity of 3D FNO for modeling global spatiotemporal evolution. This synergy enables an efficient approximation of the nonlinear operator defined in Eq. (19) and establishes a direct mapping from defect thickness distributions to full spatiotemporal wavefield.

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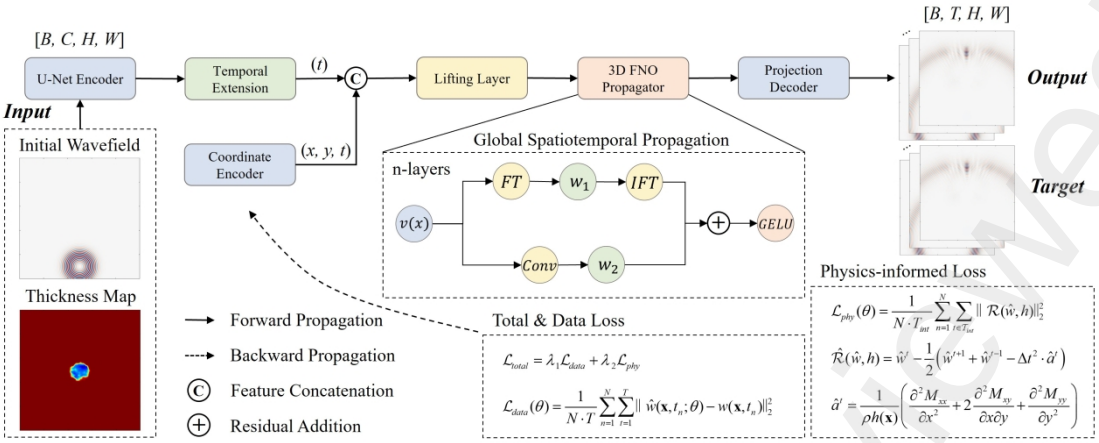


Fig. 3. Schematic of the proposed PI-DeepUFNet architecture. The framework takes the defect thickness map and initial wavefield as inputs. A U-Net encoder extracts local multi-scale geometric features, while a 3D FNO propagator models the global spatiotemporal wave evolution. The discrete KL plate equations are incorporated into the loss function to enforce dynamical consistency.

The overall architecture of the network is illustrated in Fig. 3. The input to PI-DeepUFNet is a concatenated tensor comprising the thickness field $h(\mathbf{x})$ and the initial wavefield $w(\mathbf{x}, t_0)$. This input is first processed by a geometric feature encoder based on the U-Net architecture, which extracts local geometric features and multi-scale spatial variation information through downsampling and skip connection mechanisms. The low dimensional physical space features encoded by U-Net are mapped into high dimensional latent features $v(\mathbf{x})$. Subsequently, these features are expanded along the temporal dimension via a temporal extension module, while a coordinate encoder is introduced to encode the full spatiotemporal grid coordinates (x, y, t). Following feature concatenation, a lifting layer projects the mixed features into a higher dimensional channel space, completing the embedding from the spatial domain into the spatiotemporal manifold.

The 3D FNO propagator is the core module of PI-DeepUFNet, consisting of multiple stacked Fourier operator blocks designed to learn the global propagation

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operator of waves within the medium. Unlike conventional convolutional neural networks that are limited by local receptive fields, each layer in this module incorporates a frequency-domain global convolution branch and a spatial-domain local linear branch. For the hidden feature $v_l(\mathbf{x},t)$ at the l layer, the transition to the $l+1$ layer is expressed as:

$$v_{l+1}(\mathbf{x},t)=\sigma\left(W_l v_l(\mathbf{x},t)+\mathcal{F}^{-1}\left(R_l\cdot\mathcal{F}(v_l)(\mathbf{k},\omega)\right)\right) \tag{21}$$

where, $\mathcal{F}(\cdot)$ and $\mathcal{F}^{-1}(\cdot)$ denote the Fourier transform and its inverse, respectively. R_l represents a learnable complex weight tensor in the frequency domain, W_l denotes the local linear convolution weights in the spatial domain, and $\sigma(\cdot)$ is the GELU nonlinear activation function. The frequency-domain branch efficiently captures long-range dependencies in guided wave propagation and the dispersion characteristics induced by flexural waves, while the spatial-domain branch preserves local high-frequency information and fine structures. These two components are coupled in a residual form to ensure stable modeling of the spatiotemporal evolution of guided waves.

Following propagation through multiple FNO layers, the resulting high dimensional features are mapped back to the physical space via a projection decoder, yielding the predicted out-of-plane wavefield $\hat{w}(\mathbf{x},t)$. To ensure that the predictions approximate the numerical forward modeling data while strictly satisfying the dynamical laws of guided wave propagation, a compound loss function is designed. This function comprises a data-driven term $\mathcal{L}_{data}$ and a physics-informed term $\mathcal{L}_{phy}$:

$$\mathcal{L}_{total}=\lambda_1\mathcal{L}_{data}+\lambda_2\mathcal{L}_{phy} \tag{22}$$

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where, λ_1 and λ_2 are weight hyperparameters used to balance the gradients of the two loss components. Through this dual data and physics-driven training approach, PI-DeepUFNet maintains high predictive precision while effectively suppressing the generation of non-physical solutions. This achieves efficient and reliable prediction of full spatiotemporal guided wavefield under conditions involving complex irregular defects.

3.3 Physics-informed loss function

Although PI-DeepUFNet possesses strong high-dimensional feature fitting capabilities, training exclusively on limited supervised data can still lead to overfitting or the generation of results that violate physical laws, such as abnormal wave speed dispersion or non-physical scattering artifacts. The physics-informed loss term L_{phy} is a critical design in this study, the core idea of which is to directly embed the KL control equations into the computational graph as discrete finite-difference operators, thereby imposing dynamical consistency constraints on the predicted wavefield. Unlike traditional PINNs that rely heavily on AD to calculate high-order spatiotemporal derivatives, we convert the previously constructed discrete difference operators into non-learnable deterministic convolutional layers to participate in the forward propagation of the network, which accommodates high-resolution full-field grid data while significantly reducing memory overhead.

For the predicted wavefield sequence $\hat{w}(\mathbf{x},t)$, the predicted bending moment distribution $\widehat{\boldsymbol{M}}^n(\mathbf{x})$ is first computed using the bending moment operator from Eq. (16), which subsequently yields the discrete acceleration field:

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$$\hat{a}^{n}(\mathbf{x}) = \frac{1}{\rho h(\mathbf{x})} \odot \left[\mathbf{L} * \widehat{\boldsymbol{M}}^{n}(\mathbf{x}) \right] \tag{23}$$

Based on the requirements of the explicit second-order central difference time integration scheme, any physically consistent dynamical evolution must satisfy strict recurrence constraints across any three consecutive time steps. Thus, the physical residual $\mathsf{R}(\hat{w}, h)$ is defined as the deviation between the predicted wavefield at the central time step and the theoretical value derived from the discrete dynamical equations:

$$\mathsf{R}(\hat{w}, h) = \hat{w}^{n} - \frac{1}{2}\left(\hat{w}^{n+1} + \hat{w}^{n-1} - \Delta t^{2} \cdot \hat{a}^{n} \right) \tag{24}$$

The final physics loss function is defined as the mean squared error of the residuals across all interior time steps:

$$\mathsf{L}_{phy}(\theta) = \frac{1}{N \cdot T_{int}} \sum_{n=1}^{N} \sum_{t \in \mathsf{T}_{int}} \| \mathsf{R}(\hat{w}, h) \|_{2}^{2} \tag{25}$$

By minimizing L_{phy}, the network is explicitly constrained within a solution space that conforms to the physical laws of guided wave propagation. This prior constraint not only significantly improves predictive accuracy in complex scattering regions and effectively suppresses non-physical predictions, but also fundamentally ensures the dynamical consistency and generalization capability of the model, allowing it to maintain highly stable and reliable forward performance even when encountering completely new and unknown defect topologies.

4. Numerical experiments and results

This section evaluates and analyzes the performance of the proposed method through numerical experiments. It details the dataset generation and processing,

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demonstrates the training convergence characteristics and prediction results, and quantifies the specific role of physical constraints in enhancing prediction accuracy through ablation experiments.

4.1 Dataset generation

To train and validate the PI-DeepUFNet architecture, a wavefield dataset encompassing diverse multi-scale and multi-morphology defect features was generated using the numerical forward operator defined in Eq. (19). The study focuses on a uniform aluminum plate, the material parameters of which are detailed in Table 1. The region of interest (ROI) for the numerical model is $500\text{mm} \times 500\text{mm}$. To eliminate interference from boundary reflections on the forward scattered field, the actual simulation domain extends beyond this region and is equipped with complete absorbing boundaries. To simulate irregular corrosion in engineering scenarios, thickness loss zones were randomly generated within a central 120 mm diameter area based on cloud-scan maps of actual corrosion surfaces. The maximum wall loss for each sample ranges from 0% to 50%, with equivalent diameters between 10mm and 100mm. The 3D physical layout of the simulation model is illustrated in Fig. 4(a).

Ultrasonic guided waves are excited by a vertical point force distribution applied to the top surface of the plate, with its relative coordinates in the ROI set at (250 mm, 50 mm). The excitation signal consists of a 5-cycle sine pulse modulated by a Hann window at a center frequency of 200 kHz. In numerical calculations, the spatial step is set to $dx = 1\text{mm}$, discretizing the ROI into a 500×500 grid. To satisfy the CFL stability condition, the time step is set to $dt = 0.01\mu\text{s}$, with a total simulation duration

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of $200\mu s$. Fig. 4(b) displays the thickness distribution and a total wavefield snapshot at $t = 100\mu s$ for a typical sample, clearly illustrating the complex scattering phenomena induced by irregular thickness variations.

Given the grid-independent advantages of neural operator architectures, the raw high-dimensional dataset was downsampled to accelerate model training while preserving core wavefield dynamics. Specifically, the spatial resolution was reduced to $2mm$ and the time step was extracted at intervals of $0.5\mu s$. As shown in Fig. 4(c), the downsampled signal aligns closely with the original time-domain waveform, introducing no significant numerical dissipation or phase distortion. Additionally, based on the group velocity of the $A0$ mode, a core time interval from $25\mu s$ to $125\mu s$ was selected as the effective observation window for the scattering evolution. After processing, the input for each sample consists of a $250 \times 250 \times 2$ tensor containing the 2D thickness distribution and the initial wavefield, while the corresponding target labels are full spatiotemporal wavefield tensors of size $201 \times 250 \times 250$.

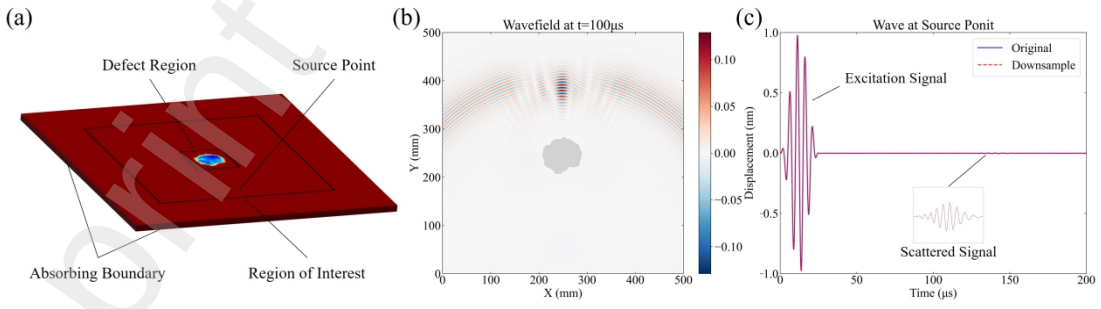


Fig. 4. Data generation and preprocessing. (a) Schematic layout of the 3D computational domain, (b) thickness distribution and full-wavefield snapshot at $t = 100us$ for a representative sample, (c) comparison of time-domain waveforms before and after downsampling.

Based on the above process, a total of 2000 independent samples were generated

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and randomly split into training, validation, and test sets at a ratio of 8:1:1. It is emphasized that the thickness distributions in the test set are entirely independent of the training data and contain novel geometric topologies never encountered during training. This ensures a strict evaluation of the generalization and dynamical extrapolation capability of PI-DeepUFNet under unknown geometric conditions.

4.2 Wavefield Prediction Results

The training and inference processes in this study were conducted on a workstation equipped with an NVIDIA RTX 4090 GPU. The Adam optimizer was employed during training with an initial learning rate of 0.001 and a total of 150 epochs. To effectively balance absolute fitting accuracy with dynamical consistency, a hybrid loss function as defined in Eq. (21) was adopted. To prevent gradient divergence caused by excessive physical constraints in the early stages of training, a dynamic weight preheating strategy was implemented. Specifically, for the first 10 epochs, the data loss weight was set to $\lambda_1 = 1$ and the physics loss weight to $\lambda_2 = 0$ for preliminary mapping based on pure data. Subsequently, both weights were adjusted to 0.5 for the remainder of the training. The optimal model was saved by monitoring the minimum fitting loss on the validation set, and all subsequent performance evaluations were conducted based on this optimal model.

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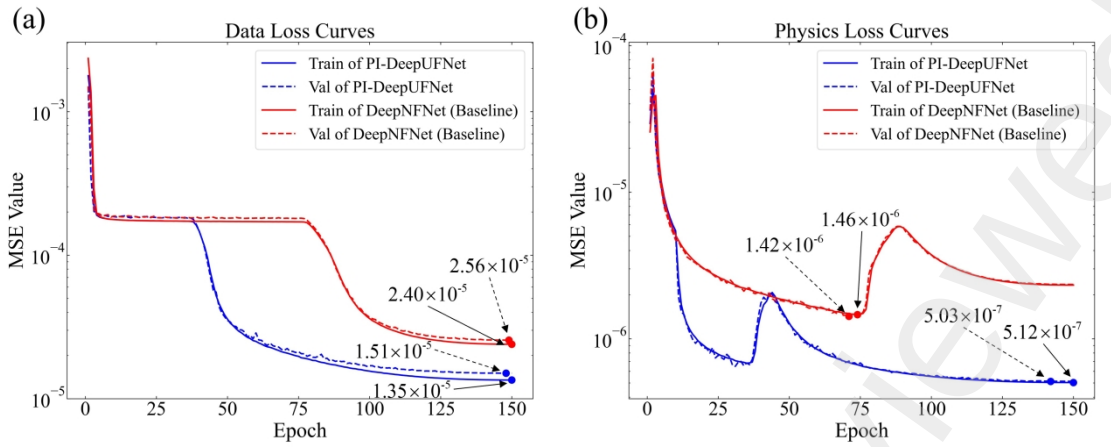


Fig. 5. Loss curves for the PI-DeepUFNet model and the baseline DeepUFNet model on the training and validation sets. (a) Data loss curves, (b) physics loss curves.

Fig. 5 compares the loss decline curves of the model on the training and validation sets. The data loss reflects the absolute fitting performance of the model relative to the ground truth wavefield, while the physics loss quantifies the degree to which the predicted wavefield adhere to physical propagation laws. Both data and physics losses decrease rapidly during the initial training phase, after which the data loss curve tends to plateau. Around epoch 37, the model escapes the local optimum due to changes in weighting and the learning rate. As training progresses, the data loss declines steadily, eventually converging to 1.51×10^{-5} on the validation set and 1.35×10^{-5} on the training set. The evolution of the physics loss exhibits more complex nonlinear dynamics, undergoing an upward jump before continuing to converge. This suggests that while data fitting and physical constraints point toward consistent optima in a global direction, their gradient descent spaces do not completely overlap, reflecting the process of the network seeking an optimal balance between the two. Final convergence values reached 5.03×10^{-7} on the validation set and 5.12×10^{-7} on the training set. Overall, the training and validation curves maintain high consistency with no significant overfitting, effectively validating the role of the hybrid loss design in

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enhancing generalization.

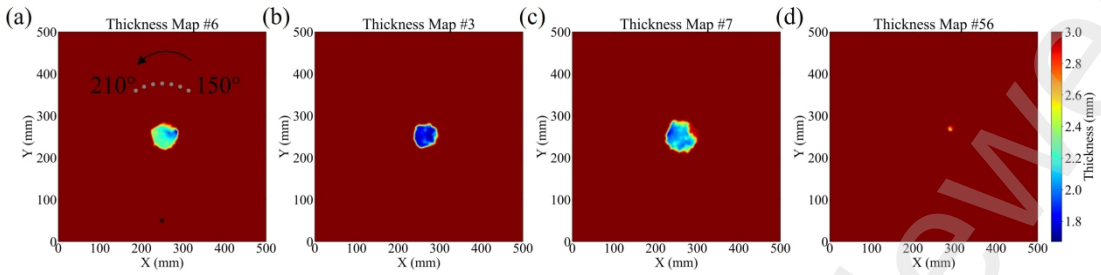


Fig. 6. Thickness distributions of four randomly selected defect samples from the test set. (a) Sample #6, (b) sample #3, (c) sample #7, (d) sample #56.

To further evaluate the ability of the model to predict scattering fields from complex defects, four representative defect samples were randomly selected from the test set, with their thickness distributions shown in Fig. 6. Taking sample #6 as an example, Fig. 7 provides a detailed comparison of full-field evolution snapshots at 60μs, 80μs, 100μs, and 120μs, including the FDM numerical ground truth, model predictions, and their residuals. The comparison shows that the predicted wavefront shape, spatial position, and amplitude distribution almost perfectly overlaps with the high-fidelity simulation results. At 60μs, as the incident wave begins to interact with the defect region and the wavefront undergoes local distortion, the model accurately captures the changes in wave speed and wavefront perturbation features near the defect. By 80μs, the guided wave gradually passes through the defect region, and the model successfully predicts the generation of scattered waves and the formation of multiple scattering ripples. As time progresses to 100μs and 120μs, distinct scattered waves and interference structures appear, and the model continues to reconstruct the complex wavefront structures and amplitude distributions with high-fidelity. As shown in the residual maps in Fig. 7(c), errors are primarily concentrated in the strong scattering regions near the defects, with peak errors not exceeding 13% and the global average

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error remaining stable below 1%. This intuitively demonstrates that the neural operator possesses exceptionally high prediction fidelity across the entire field.

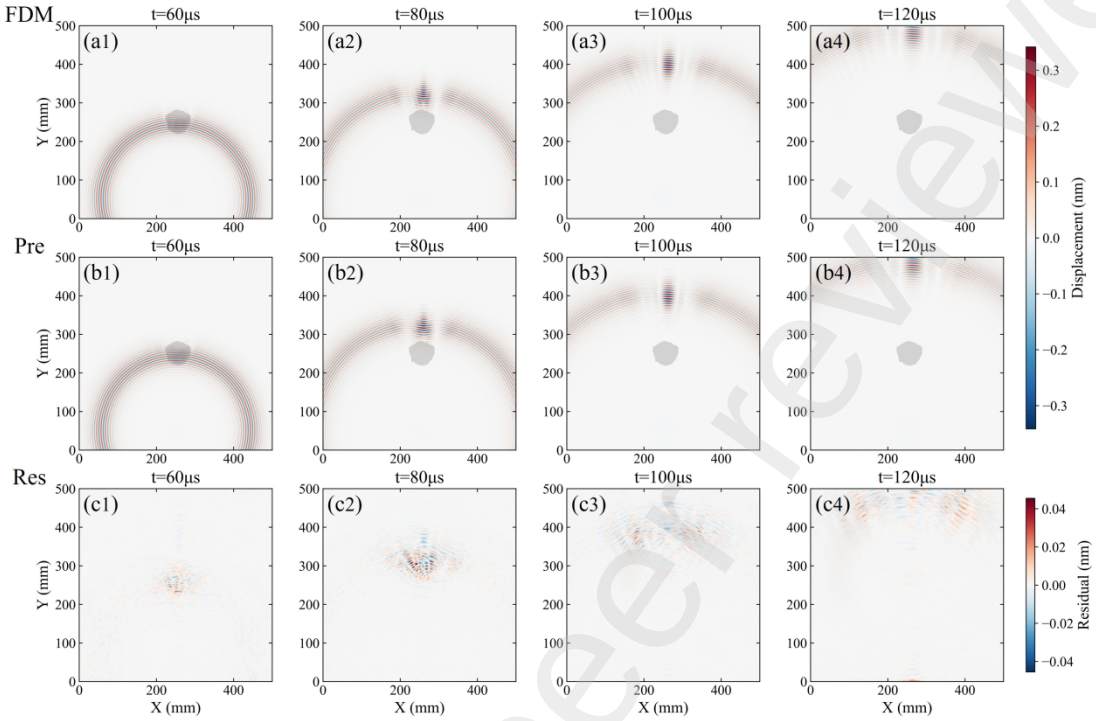


Fig. 7. Wavefield snapshots for defect sample #6 at different time steps. The rows from top to bottom correspond to the FDM ground truth, PI-DeepUFNet predictions, and absolute residuals, respectively.

To quantitatively evaluate the performance of the proposed PI-DeepUFNet, we employ three standard metrics with the relative L2 error (RL2) to measure overall global accuracy, the root mean square error (RMSE) for absolute deviation, and the Pearson correlation coefficient (PCC) to assess waveform phase consistency and shape similarity. Detailed calculation results are listed in Table 2, where sample #6 achieved RMSE, RL2, and PCC values of 1.97×10^{-3}, 4.46×10^{-2}, and 0.9990, respectively. Comprehensive statistics for the first 100 samples in the test set yield an average RMSE of 3.08×10^{-3}, an average RL2 of 6.98×10^{-2}, and an average PCC as high as 0.9971. This near-unity PCC indicates that the predicted waveforms are highly consistent with the labels in terms of evolution trends and amplitude variations, proving from a physical

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perspective that the model has accurately internalized the complex dispersion attributes of guided waves. This capacity for precise capture of phase features enables the model to reliably identify subtle scattering ripples caused by irregular defects, thereby laying a solid data foundation for high-resolution imaging and quantitative evaluation based on time of flight.

Table 2
Quantitative evaluation of the wavefield predictions generated by PI-DeepUFNet.

Sample	PI-DeepUFNet		
	RMSE	RL2	PCC
Test set			
# 3	5.20×10^{-3}	1.18×10^{-1}	0.9931
# 6	1.97×10^{-3}	4.46×10^{-2}	0.9990
# 7	4.52×10^{-3}	1.02×10^{-1}	0.9948
# 56	8.45×10^{-4}	1.92×10^{-2}	0.9998
Avg (0~100)	3.08×10^{-3}	6.98×10^{-2}	0.9971
Experiment			
# 1	1.73×10^{-3}	3.93×10^{-2}	0.9992

4.3 Ablation Study: Impact of physics-informed constraint

To quantitatively evaluate the specific contribution of physics constraints, an ablation study was conducted. A purely data-driven neural network without the PDE residual loss term is defined as the DeepUFNet baseline. During training of the baseline model, the PDE residual serves only as a monitoring metric and does not participate in backpropagation. As shown in the loss curves of Fig. 5, the convergence of the model without physics constraints is significantly slower, with final data and physics losses stabilizing at approximately 1.7 and 2.8 times those of the PI-DeepUFNet model respectively. Notably, after a jump in the data loss curve of the baseline model near epoch 75, the physics loss fails to return to its previous low point despite a continued smooth decline in data loss. This suggests that while the purely data-driven model

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achieves acceptable fitting within the training distribution, its mapping relationship does not strictly follow the underlying wave dynamics, leading to energy divergence or phase misalignment during extrapolation. In contrast, the physics guided mechanism constrains the network to converge toward optimal solutions that strictly satisfy the KL dynamical equations within the hypothesis space, demonstrating superior convergence stability and energy conservation.

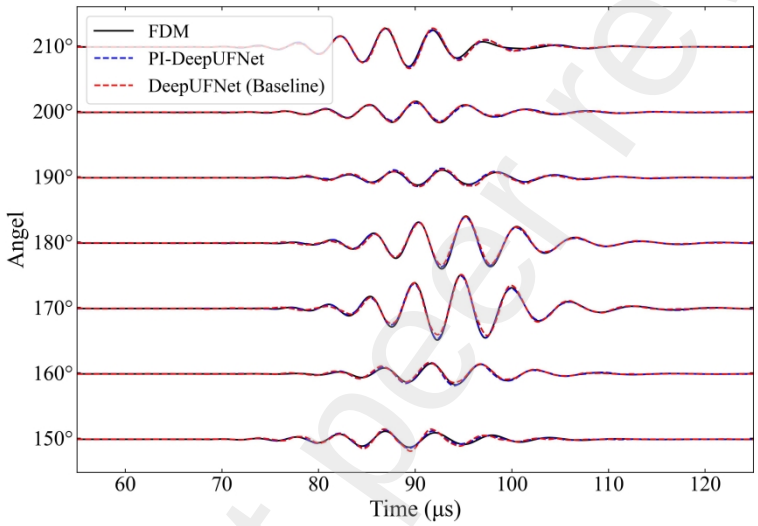


Fig. 8. Time domain waveforms at observation points for sample #6 including the FDM benchmark, PI-DeepUFNet predictions, and the baseline DeepUFNet results.

To verify this conclusion regarding local details, seven observation points were uniformly selected along the forward scattering directions from 150° to 210° in the shadow region behind the defect of sample #6 as shown in Fig. 6(a). As illustrated in Fig. 8, the black solid lines represent the FDM simulation benchmark, while the blue and red dashed lines represent the predictions of PI-DeepUFNet and the baseline model respectively. The results show that the waveforms predicted by PI-DeepUFNet overlap almost perfectly with the FDM benchmark, maintaining high consistency in wave packet phase, amplitude, and interference patterns. However, the baseline model without physics constraints exhibits varying degrees of distortion as the observation

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angle changes. This error is significantly concentrated in the middle and late stages of wave packet evolution. The physical root of this discrepancy lies in the failure of the purely data-driven network to fully internalize the coupling between spatial thickness distribution and guided wave dispersion characteristics, resulting in numerical dissipation and phase shifts during complex transmission and diffraction processes.

To further compare the effect of physics constraints across the entire spatial domain, the scattered wavefield of four test samples at $80\mu s$ were extracted. To highlight the scattering features caused by defects by eliminating the masking effect of the strong background incident field, the scattered wavefield w_{scat} is defined as:

$$w_{scat}(\mathbf{x},t) = w_{total}(\mathbf{x},t) - w_{base}(\mathbf{x},t) \tag{26}$$

where, w_{base} is the reference wavefield in a uniform plate without defects.

The comparison results in Fig. 9 indicate that since corrosion defects belong to typical thickness reduction geometries, their overall reflection response is weak and the wavefield is dominated by transmission and forward scattering behavior. For sample #6, both models reproduce the scattered wavefield distribution relatively well, but PI-DeepUFNet offers higher fidelity with PCC values reaching 0.9954 and 0.9878 respectively. For sample #3, which features severe thickness reduction and complex scattering, the wavefield predicted by the baseline model shows visible structural distortion, with the PCC value dropping from 0.9576 to 0.8988. For the large defect in sample #7 and the small defect in sample #56, the PI-DeepUFNet predictions also remain closer to the FDM overall. Particularly in sample #56, DeepUFNet shows significant local distortion while PI-DeepUFNet maintains good reconstruction of the

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scattered field structure. Corresponding PCC values improved from 0.9274 and 0.8626 to 0.9720 and 0.9105.

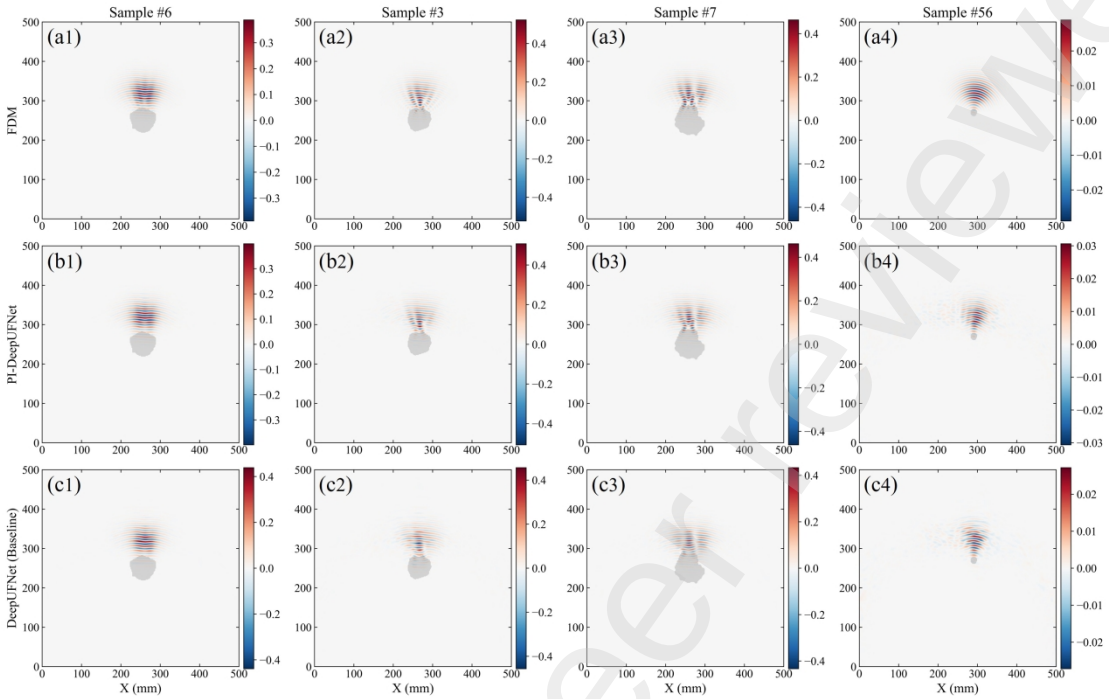


Fig. 9. Scattered wavefield for various defect samples at $t = 80\mu s$, where rows from top to bottom correspond to the FDM numerical benchmark, PI-DeepUFNet predictions, and the baseline DeepUFNet results.

Beyond the qualitative comparison of waveforms and phase indicative PCC, we further calculated RMSE and RL2 to quantitatively evaluate the scattered field predictions for the first 100 samples in the test set at $80\mu s$ as listed in Table 3. Results indicate that PI-DeepUFNet outperforms the purely data-driven baseline model across all representative samples and in terms of the overall average. The average results for the test set show that the RMSE, RL2, and PCC for PI-DeepUFNet are 4.03×10^{-3}, 2.59×10^{-1}, and 0.9589 respectively, while the corresponding metrics for DeepUFNet are 6.28×10^{-3}, 3.74×10^{-1}, and 0.9191.

In summary, the systematic ablation experiments and multi-dimensional quantitative evaluations confirm the scientific validity of embedding the KL dynamical

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partial differential operator into the neural network. Guided by prior physical knowledge, the neural network is no longer limited to fitting shallow data mapping relationships but learns the physical laws of guided wave propagation and dispersion at the mechanistic level. This deep integration of physics and data fundamentally suppresses the non-physical artifacts common in black box deep learning models. It not only grants the model exceptional generalization capability when facing unknown complex defects but also establishes a reliable physical foundation for high resolution ultrasonic NDT and imaging based on deep learning.

Table 3

Quantitative evaluation of the predicted scattered wavefield at $t = 80\mu s$ for the proposed PI-DeepUFNet and the baseline DeepUFNet.

Sample	PI-DeepUFNet			DeepUFNet (Baseline)		
	RMSE	RL2	PCC	RMSE	RL2	PCC
Test set						
# 3	6.91×10^{-3}	2.89×10^{-1}	0.9576	1.05×10^{-2}	4.41×10^{-1}	0.8988
# 6	2.54×10^{-3}	9.90×10^{-2}	0.9954	4.18×10^{-3}	1.63×10^{-1}	0.9878
# 7	5.94×10^{-3}	2.35×10^{-1}	0.9720	9.46×10^{-3}	3.75×10^{-1}	0.9274
# 56	7.51×10^{-4}	4.16×10^{-1}	0.9105	9.31×10^{-3}	5.16×10^{-1}	0.8626
Avg (0~100)	4.03×10^{-3}	2.59×10^{-1}	0.9589	6.28×10^{-3}	3.74×10^{-1}	0.9191

5. Discussion

This section further discusses the predictive performance of the PI-DeepUFNet model with a focus on evaluating how geometric features such as defect size and depth influence model accuracy. Additionally, scanning experiments are designed to validate the actual representation capability of the model.

5.1 Sensitivity to defect morphology

In practical ultrasonic NDT, the geometric topology of corrosion defects exhibits a high degree of randomness. As described by the Kirchhoff-Helmholtz integral

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equation, the distribution characteristics of the forward scattered field depend strictly on the geometric shape of the scattering boundary and the perturbation of local flexural rigidity:

$$u_{sc}(\mathbf{R}) = \int_{S(\mathbf{r})} \left[g(\mathbf{R}-\mathbf{r}) \cdot \frac{\partial u_{inc}(\mathbf{r})}{\partial n} - u_{inc}(\mathbf{r}) \cdot \frac{\partial g(\mathbf{R}-\mathbf{r})}{\partial n} \right] dS(\mathbf{r}) \tag{27}$$

where, u_{sc} represents the scattered field, g is the Green's function, and u_{inc} is the incident field. To systematically investigate the sensitivity of the PI-DeepUFNet model to geometric features, we generated an additional independent test set containing 500 irregular defect samples and calculated various evaluation metrics for the scattered field predicted at $80\mu s$.

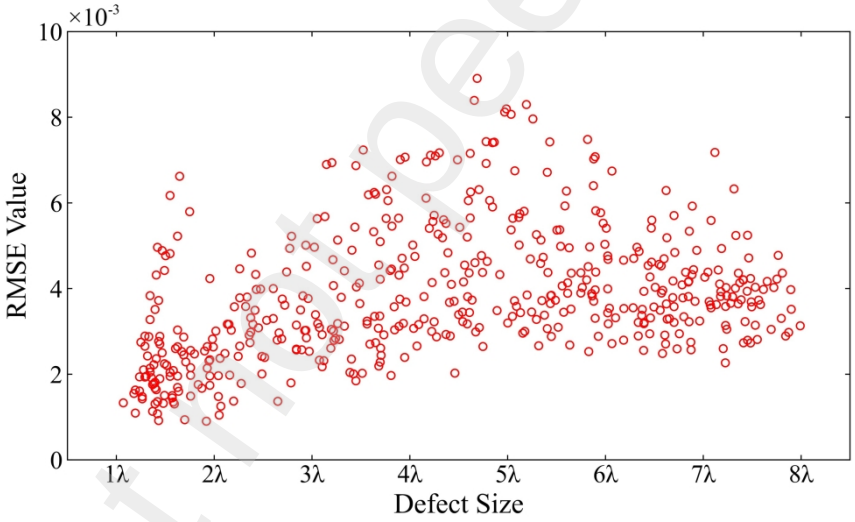


Fig. 10. Impact of defect size on PI-DeepUFNet prediction accuracy.

Fig. 10 intuitively reflects the impact of equivalent defect diameter on the predicted RMSE. The horizontal axis represents the defect size relative to the center frequency wavelength λ ranging from 1.07λ to 7.99λ. It can be observed that when the defect size is small, its acoustic response resembles point scattering and the wavefield distortion is relatively simple. The model exhibits extremely low prediction errors with RMSE concentrated around 2×10^{-3}. As the defect size increases, the

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prediction error shows a slow upward trend due to the increase in scattering boundary length and the intensification of internal multi-interference effects. After the defect size exceeds approximately 2.5λ, the degradation of accuracy due to defect size gradually plateaus. The RMSE no longer continues to diverge but fluctuates within a narrow range around 4×10^{-3}. This phenomenon indicates that PI-DeepUFNet possesses excellent representation capability for large defects. Its global FNO has successfully learned macro-scattering physical patterns and does not suffer from significant performance degradation as geometric scales increase.

Compared to the lateral size, the longitudinal depth of the defect has a more direct nonlinear impact on the local wave speed and energy evolution of guided waves. Fig. 11 illustrates the relationship between wall loss and predicted RMSE which exhibits clear piecewise evolution characteristics. When the thickness loss is below 21%, the perturbation of local stiffness is weak and the model exhibits extreme robustness with RMSE stably distributed at a low level of 1×10^{-3} and not deteriorating with increasing depth. In the 21% to 28% range, RMSE enters a transition zone with an upward trend as local frequency dispersion effects intensify. When the wall loss further exceeds 28% up to 50%, the overall error center continues to rise but the distribution of data points shows strong discreteness. This suggests that in the deep corrosion stage, wall loss is no longer the sole dominant factor controlling prediction error, as high-order geometric features such as complex edge contours of the defect begin to significantly interfere with the scattering process.

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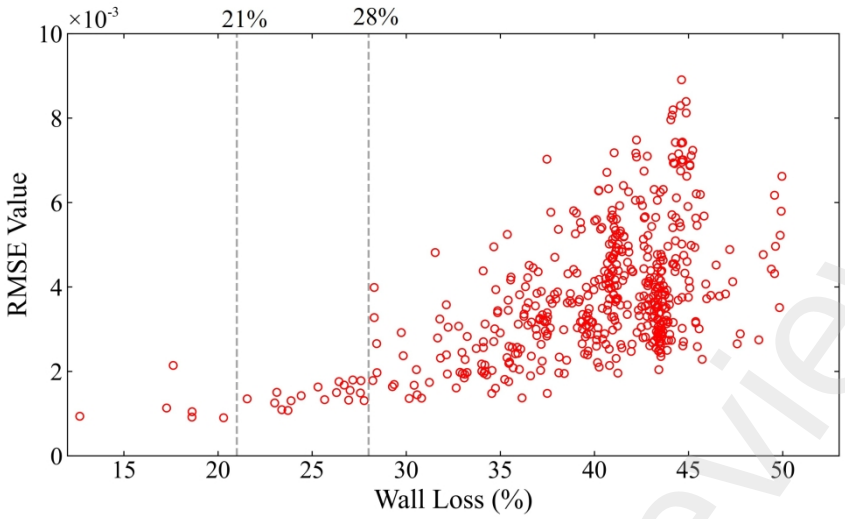


Fig. 11. Impact of maximum wall loss on PI-DeepUFNet prediction accuracy.

Overall, although the wavefield prediction accuracy of PI-DeepUFNet is influenced to some extent by physical parameters such as defect size and depth, its comprehensive performance remains at a high-fidelity level. In the test set of 500 independent samples, the average RMSE of the scattered wavefield at 80μs is only 3.81×10^{-3} and the average RL2 error is 9.51×10^{-2}. Notably, the PCC representing waveform phase accuracy is distributed between 0.9751 and 0.9997 with an average of 0.9947 and 88.2% of the samples exceeding 0.99. These statistical results indicate that with the help of deep guidance from physics differential equations, PI-DeepUFNet not only accurately maps the evolution representation of the spatiotemporal wavefield but also internalizes the underlying mechanical mechanisms governing guided wave propagation and scattering within a broad geometric parameter space. This establishes a solid physical foundation for quantitative health monitoring of thin-walled structures.

5.2 Experimental validation

To further validate the effectiveness of the proposed PI-DeepUFNet model in a real physical environment, an ultrasonic wave measurement platform was established

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based on a SLDV. The specimen is a $1000\times1000\times3\text{mm}$ uniform aluminum plate with material and mechanical parameters identical to those used in the numerical simulation. To avoid interference from boundary reflections, a $500\times500\text{mm}$ area in the center of the plate was designated as the effective observation region. A piezoelectric (PZT) interdigital transducer was bonded at coordinates of $(250\text{mm},50\text{mm})$ using epoxy resin to serve as the guided wave excitation source. An observation circle with a radius of 127 mm was centered on the research area to capture wavefield characteristics as shown in Fig. 12(a).

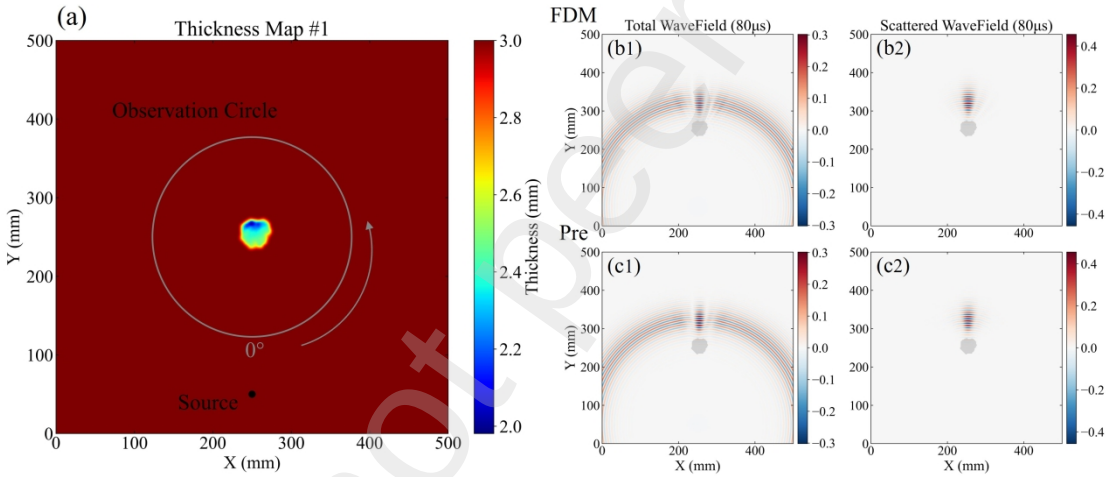


Fig. 12. (a) Schematic of the sample thickness map and observation circle setup, (b) FDM benchmarks for total and scattered wavefield at $80\mu\text{s}$, (c) PI-DeepUFNet prediction results.

Before introducing experimental data, the total wavefield and scattered wavefield of PI-DeepUFNet and the FDM benchmark at $80\mu\text{s}$ were first compared. As shown in the visualization results of Fig. 12(b) and Fig. 12(c), PI-DeepUFNet maintains high consistency with the FDM benchmark for both the main wavefront in the total wavefield and the local perturbations in the scattered wavefield. For this sample, the RMSE, RL2, and PCC of the total wavefield are 2.10×10^{-3}, 5.24×10^{-2}, and 0.9986 respectively, while the corresponding metrics for the scattered wavefield are

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2.13×10^{-3}, 1.07×10^{-1}, and 0.9942.

During the actual SLDV measurement phase, an irregular defect with a topology identical to the simulation model was machined on the aluminum plate. Considering machining feasibility, the defect was depth approximated using a step-milling process. The experiment utilized a 5-cycle sine pulse at 200 kHz and 360 observation points were uniformly distributed along the observation circle. The lowest point of the circle is defined as 0° and signals were collected counterclockwise at each observation point. The overall experimental setup is illustrated in Fig. 13(a). The first arrival wave packet was extracted for each observation point and the peak-to-peak amplitude was calculated as the response magnitude for that direction. All observed values were then normalized. To ensure comparability among the three, an identical signal processing flow was applied to the FDM benchmark and PI-DeepUFNet prediction results to extract the angular magnitude distribution along the corresponding circle.

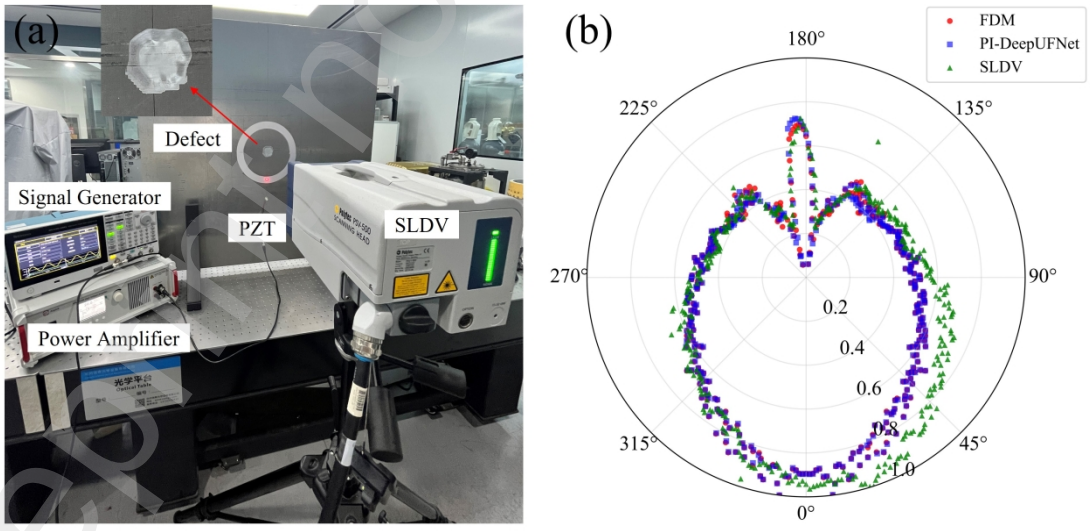


Fig. 13. Experimental setup and results. (a) SLDV experimental platform, (b) Comparison of normalized amplitude directivity patterns from FDM simulation, PI-DeepUFNet prediction, and SLDV measurement.

The polar plot in Fig. 13(b) characterizes the directional distribution of wavefield

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energy in space. Red circles represent the FDM benchmark, blue squares represent the PI-DeepUFNet prediction, and green triangles represent the actual SLDV measurement results. It can be observed that the three results exhibit strong correlation in overall distribution trends. In the main lobe direction near 0°, all three exhibit the highest response. The magnitude gradually decays as the angle deviates from the main propagation direction and a local peak caused by backscattering from the defect appears near 185°. This indicates that PI-DeepUFNet can accurately capture the main mechanisms governing the angular scattering characteristics of irregular defects. Further comparison reveals that PI-DeepUFNet and FDM almost overlap across most angles, suggesting that the model has accurately inherited the angular scattering characteristics of the numerical forward model. In contrast, although the overall trend of the SLDV measurement results is consistent, fluctuations in local peak values and positions occur within the 0° to 90° range and near 185°. These errors primarily stem from machining tolerances of the defect, coupling layer effects of the PZT, and fluctuations in the laser reflection signal under non-steady environments. Nevertheless, the high degree of consistency in the overall envelope fully validates the predictability of the model under real and complex physical boundaries.

In summary, PI-DeepUFNet demonstrates stable cross-domain prediction capability. Compared to traditional numerical simulations that are extremely time consuming, this physics guided model can complete the forward inference of spatiotemporal wavefield for complex irregular defects within seconds. This allows it to serve as an efficient real-time forward operator embedded into guided wave based

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intelligent quantitative inversion and imaging algorithms. It can also provide a reliable calculation engine for SHM digital twin wavefield libraries.

6. Conclusion

This study proposes a PI-DeepUFNet framework merging KL plate theory with physics-informed neural operators to address the rapid forward prediction of full spatiotemporal wavefield in plates containing irregular defects. By employing the fourth-order flexural wave dynamic equations under spatially variable thickness as physical priors, the proposed method integrates the local geometric feature extraction capabilities of U-Net with the global spatiotemporal evolution modeling of 3D FNO. This synergy establishes a robust mapping from defect thickness distributions to the resulting guided wave responses.

Quantitative results demonstrate that PI-DeepUFNet can accurately predict guided wave wavefield under complex defect conditions. Within the test set, the average RMSE, RL2, and PCC for wavefield predictions on unknown defect samples reached 3.08×10^{-3}, 6.98×10^{-2} and 0.9971 respectively. Ablation analysis reveals that the embedding of physical constraints addresses the generalization bottlenecks of purely data-driven models, enhancing prediction accuracy and stability in complex scattering. Sensitivity analysis regarding defect size and depth features, along with cross validation through SLDV scanning experiments, further confirms the robustness and applicability of the model in real physical scenarios.

Compared to traditional numerical methods, this framework significantly improves forward modeling efficiency while maintaining high-fidelity, providing a

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computational foundation for SHM digital twins and subsequent quantitative inversion based on guided waves. Furthermore, the forward modeling paradigm established in this work possesses excellent theoretical portability. By replacing the discrete physical operators in the loss function with the governing equations of other target systems, this scheme can be extended to other forward modeling tasks involving diverse physical fields. This strategy provides a universal approach for the integration of physical priors with operator learning.

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